



On elliptical multilevel models

Roberto F. Manghi, Gilberto A. Paula & Francisco José A. Cysneiros

To cite this article: Roberto F. Manghi, Gilberto A. Paula & Francisco José A. Cysneiros (2016): On elliptical multilevel models, Journal of Applied Statistics, DOI: [10.1080/02664763.2015.1134445](https://doi.org/10.1080/02664763.2015.1134445)

To link to this article: <http://dx.doi.org/10.1080/02664763.2015.1134445>



Published online: 18 Jan 2016.



Submit your article to this journal [↗](#)



Article views: 11



View related articles [↗](#)



View Crossmark data [↗](#)

On elliptical multilevel models

Roberto F. Manghi^a, Gilberto A. Paula^b and Francisco José A. Cysneiros^a

^aDepartamento de Estatística, Universidade Federal de Pernambuco, Brazil; ^bInstituto de Matemática e Estatística, Universidade de São Paulo, São Paulo, Brazil

ABSTRACT

Multilevel models have been widely applied to analyze data sets which present some hierarchical structure. In this paper we propose a generalization of the normal multilevel models, named elliptical multilevel models. This proposal suggests the use of distributions in the elliptical class, thus involving all symmetric continuous distributions, including the normal distribution as a particular case. Elliptical distributions may have lighter or heavier tails than the normal ones. In the case of normal error models with the presence of outlying observations, heavy-tailed error models may be applied to accommodate such observations. In particular, we discuss some aspects of the elliptical multilevel models, such as maximum likelihood estimation and residual analysis to assess features related to the fitting and the model assumptions. Finally, two motivating examples analyzed under normal multilevel models are reanalyzed under Student-*t* and power exponential multilevel models. Comparisons with the normal multilevel model are performed by using residual analysis.

ARTICLE HISTORY

Received 20 October 2014
Revised 21 October 2015
Accepted 16 December 2015

KEYWORDS

Correlated data; elliptical models; multilevel models; multivariate Student-*t* distribution; robust estimation

AMS SUBJECT CLASSIFICATION

62J05; 62J20

1. Introduction

Multilevel modeling is an important and widely applied technique in statistical modeling for analyzing data sets which present a hierarchical, nested, or clustered structure. More specifically, such hierarchy may be viewed as the existence of units grouped in different levels. For instance, researches that provide data collected in the humanities and biological sciences generally have a hierarchical frame. In the area of inheritance involving animals or people, there is a natural arrangement of units on levels, where offsprings are grouped within families. Individuals from the same parents tend to have more similarities in their physical and mental characteristics than individuals chosen randomly from the population as a whole. Children of the same family, as an example of this case, tend to have some similarities with their parents concerned on a group of specific physical aspects.

The multilevel model references comprise, for example, Raudenbush and Bryk [28] and West *et al.* [32], but before the proposal of multilevel modeling, an important class to be considered is the variance component models, as seen in Searle *et al.* [30]. In this class, the particular structure of variance and covariance may be modeled by the determination of

random effects relative to certain factors presented in the study. Harville [12] derives some maximum likelihood approaches for the variance component estimation.

Generalizations of the variance component models have been proposed, such as the normal linear mixed models (see Laird and Ware [18]), efficiently applied in longitudinal and repeated-measurement studies. McLean *et al.* [23] compared results of applications involving variance analysis models by using mixed models. Considering this later class of models, unbalanced data sets motivated the use of iterative methods for the parameter estimation and the EM (Expectation-Maximization) algorithm proposed by Dempster *et al.* [3] has been largely applied. This algorithm is also used for obtaining the variance component estimates, as described, for instance, in Dempster *et al.* [4] and Laird [17].

Most of the techniques developed for mixed models may be used in multilevel models. There are studies that detail specific methods for the parameter estimation, hypothesis testing and some procedures on residual analysis and diagnostics in multilevel models under normal errors. For example, West *et al.* [32] discussed the application of multilevel models by using various examples involving data sets with two or three levels of hierarchy. In Hilden-Minton [13] one may see some diagnostic techniques for such models, whereas in Klocke *et al.* [15] the focus is on the use of rank-based methods for the problem of dealing with outliers in such models.

An important feature of normal mixed models is the lack of robustness of the maximum likelihood estimates under the presence of outlying observations, which tend to exercise large influence on the parameter estimates and inference. A proposal for dealing with this kind of problem is the use of error distributions with heavier tails than the normal ones, such as Student-*t* models, in order to better accommodation of outlying observations. Substantial work has been available on Student-*t* models. For example, Kowalsky *et al.* [16] compared Bayesian and classical techniques in multivariate Student-*t* models. Fernández and Steel [7] illustrated some pitfalls encountered in Bayesian and maximum likelihood methods under Student-*t* regression models with unknown degrees of freedom. Lange *et al.* [19] addressed the robust aspects of the maximum likelihood estimates in Student-*t* models for longitudinal data with fixed or unknown degrees of freedom. Liu and Rubin [22] presented extensions of the EM algorithm for the estimates in models with multivariate Student-*t* distribution, whereas Pinheiro *et al.* [27] applied extensions of the EM algorithm for the robust parameter estimation in mixed models with Student-*t* error distribution. Little [21] discussed the robust estimation in models with missing data and Cysneiros and Paula [2] derived the use of one-sided hypothesis testing in regression models with multivariate Student-*t* distribution.

The elliptical class contains all symmetric continuous distributions with light- and heavy-tailed distributions than the normal ones. We may found in this class, among others, the normal distribution as a particular case as well as distributions with heavier (Student-*t*, slash, logistic II), lighter (logistic I) and lighter or heavier (power exponential) tails than the normal distribution. The selection of an appropriate error distribution in the elliptical class may be performed, for instance, by goodness-of-fit measures, residual analysis and sensitivity studies. Various works have been published on linear mixed models under elliptical errors. For example, Galea *et al.* [8] developed local influence analysis in multivariate elliptical models, whereas Galea *et al.* [9] addressed diagnostic techniques for the univariate case. Lindsey [20] discussed the application of the power exponential models in repeated-measurement studies in which the response presents shorter or longer tails, whereas Savalli

et al. [29] proposed the use of a score-type test to assess variance and covariance components in elliptical mixed models and Osorio *et al.* [25] developed local influence analysis in this class of models with emphasis in longitudinal data structure. Recently, Ibacache-Pulgar *et al.* [14] extended such techniques for semiparametric elliptical linear mixed models.

The aim of this paper is to propose a generalization of the normal multilevel models, named elliptical multilevel models, in the sense of allowing light- and heavy-tailed error distributions in the hierarchy levels. The paper is organized as follows. In Section 2 two motivating examples are presented in which the lack of robustness of the maximum likelihood estimates under normal multilevel models is evidenced. In Section 3 the elliptical class of distributions is presented as well as some of its properties. The elliptical multilevel models are defined in Section 4, and in particular some aspects of the maximum likelihood estimation are discussed. Standardized marginal and conditional residuals are derived in Section 5. Section 6 presents a small simulation study to assess the empirical distribution of the maximum likelihood estimates for moderate and large sample sizes. The motivating examples are reanalyzed in Section 7 under elliptical multilevel models and a comparison with the normal model is performed by using residual analysis. Section 8 deals with some conclusions on the results obtained in Section 7 and on the advantages of using elliptical multilevel models in data analysis.

2. Motivating examples

2.1. The rat pup study

For the proposal of motivation we will consider as the first example the data set presented in Pinheiro and Bates [26], the Rat Pup study, in which 30 female were randomly assigned to receive one of the three doses (low, high or control) of an experimental compound. The aim of the study was to compare the weight of newborn rats in litters that received high or low dosages with the weight of rat pups whose litters received the control treatment. One has 10 rats randomized to receive each one of the three treatments, but three females that received high-dose died, and for these cases no data for their respective litters was considered. Then, one has an unbalanced study in which the number of litters per treatment and the number of rat pups per litter are not constant for the three treatments.

In addition, the data present two levels of hierarchy. Each female rat was randomized to receive one of the three treatments, and each rat pup belongs to only one litter. We say in this case that the rat pups constitute the units of level 1, and so they are nested in units of level 2 (litters considered in the study). Weights of the pups from the same litter may exhibit correlation. Thus, it may be reasonable to consider a random effect for each litter, as well as fixed effects for analyzing variations between treatments. For the 27 litters we have a total of 322 rat pups in this application.

Considering the litter size, that is, the number of pups born from each one of the 27 females, we may see from Figure 1 that the median value of the pup weight in general decreases as the number of pups increases in the litters, except for the group that received the low-dose treatment. For the normal multilevel model fitting we will consider the same structure given in West *et al.* [32]. These authors proposed the model

Level 1 equation:

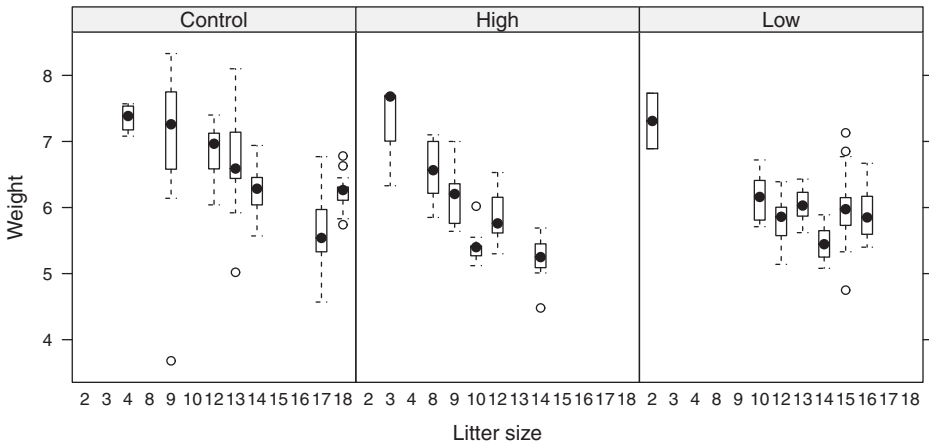


Figure 1. Boxplots of the rat pup weight by litter size for the three treatments considered in the Rat Pup study.

$$\text{Weight}_{jk} = \gamma_{1k} + \gamma_{2k}\text{Sex}_{jk} + \epsilon_{jk};$$

Level 2 equations:

(1)

$$\gamma_{1k} = \beta_{11} + \beta_{12}\text{Treat1}_k + \beta_{13}\text{Treat2}_k + \beta_{14}\text{Litsize}_k + b_{1k}; \quad \gamma_{2k} = \beta_{21},$$

for $j = 1, 2, \dots, n_k$ and $k = 1, 2, \dots, 27$, where n_k is the number of pups belonging to the k th litter totaling $K = 27$ litters. Here, Weight_{jk} denotes the weight (in grams) of the j th pup in the k th litter, whereas Sex_{jk} (1, if the j th rat pup of the k th litter is female, or 0, if the j th rat pup of the k th litter is male), Treat1_k (1, if the k th litter received the high dosage, or 0, otherwise) and Treat2_k (1, if the k th litter received the low dosage, or 0, otherwise) are dichotomous variables. The quantity Litsize_k denotes the size (number of pups) of the k th litter. In the first equation, regarding level 1 of the model, we have γ_{1k} as the intercept for the k th litter, γ_{2k} as the parameter associated with the sex of the pups belonging to the k th litter and a random error ϵ_{jk} associated with the weight. It is assumed that

$$\epsilon_{jk} \stackrel{\text{iid}}{\sim} \begin{cases} N(0, \sigma_{\text{CT}}^2), & \text{if the } k\text{th litter received the control dosage;} \\ N(0, \sigma_{\text{HL}}^2), & \text{if the } k\text{th litter received the high or low dosage.} \end{cases}$$

Also, one has $b_{1k} \stackrel{\text{iid}}{\sim} N(0, \sigma_{\text{L}}^2)$ and independence among b_{1k} and ϵ_{jk} , for $j = 1, 2, \dots, n_k$ and $k = 1, 2, \dots, 27$.

In the first equation for the level 2 model, we have β_{11} as the overall average for the intercepts γ_{1k} for litters containing no puppies ($\text{Litsize}_k = 0$) that received the control treatment, β_{12} representing the effect on the intercept γ_{1k} for the litters that received the high-dose treatment, β_{13} the effect on γ_{1k} for the litters under the low-dose treatment and β_{14} being the parameter associated with the litter size. We still have, in this equation, a random effect related to γ_{1k} , denoted by b_{1k} , for $k = 1, 2, \dots, 27$. The parameter β_{21} , in the second equation for the level 2, indicates that the effect of the variable Sex on the expected

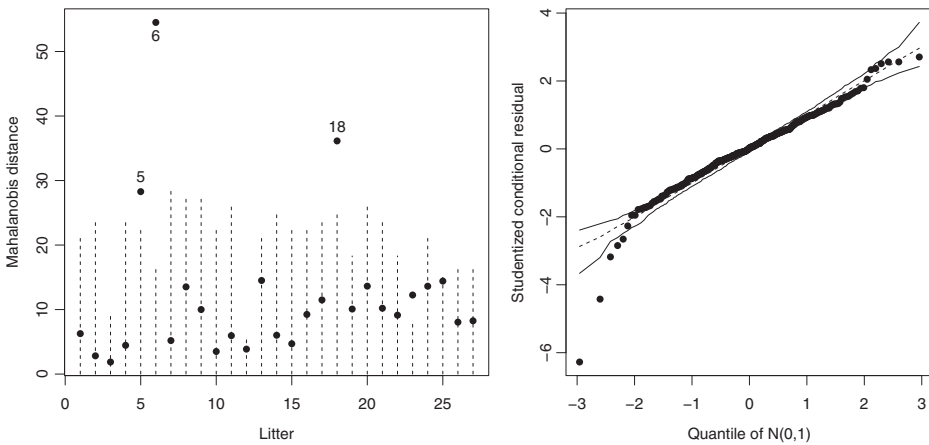


Figure 2. Estimated Mahalanobis distances with their respective 95% quantiles by litter (left) and normal probability plot with 95% simulated confidence band for the studentized conditional residual (right) under the normal multilevel model fitted to the Rat Pup data.

weight of the pups is the same for all the litters. The model (1) may be expressed as

$$\text{Weight}_{jk} = \beta_{11} + \beta_{12}\text{Treat1}_k + \beta_{13}\text{Treat2}_k + \beta_{14}\text{Litsize}_k + \beta_{21}\text{Sex}_{jk} + b_{1k} + \epsilon_{jk},$$

for $j = 1, 2, \dots, n_k$ and $k = 1, 2, \dots, 27$. Here, one has that β_{11} is the model intercept, β_{12} is the fixed-effect on the expected value of the pup weight in the litters that received the high-dose treatment, β_{13} is the fixed-effect on the expected weight of the pups that received the low-dose treatment, β_{14} represents the fixed-effect of the litter size on the expected weight of the rats and β_{21} is the fixed-effect on the expected weight of the female rat pups. We still have that the b_{1k} 's are random effects of the litters on the newborn rat weight and the ϵ_{jk} 's are random errors associated with the values of the respective rat weights considered in this study.

Concerning the residual analysis of the model (1) fitting, the points in the left graph of Figure 2 represent the estimates of the Mahalanobis distance for each litter and the top of each dashed line segment represents the 95% quantile value of a chi-squared distribution with n_k degrees of freedom, for $k = 1, 2, \dots, 27$. Under the model (1) the Mahalanobis distance follows a chi-squared distribution with n_k degrees of freedom (see Lange *et al.* [19]). Points above the upper ends of the respective line segment may indicate outstanding litters. Litters 5, 6 and 18 present higher values for the estimated Mahalanobis distance than their respective quantiles. Thus, such litters may be considered outstanding. The average weight of the rat pups for these litters are 6.89g, 6.97g and 6.14g, respectively. Litters 5 and 6 received the control dosage of the experimental compound, whereas litter 18 was subjected to the low-dose treatment. From the normal probability plot with 95% simulated confidence band for the studentized marginal residual (omitted here), no unusual departure is observed, the majority of the points lying within the simulated confidence band. There is a point away from the simulated confidence band, possibly related to the litter 6. However, from the right graph in Figure 2, there are several points outside the confidence band, and therefore the assumption of normality for the conditional errors seems to be not

satisfied. Thus, the presence of outlying litters is a motivation for the use of distributions with heavier tails than the normal ones.

2.2. Gross state product data

In the second motivating example we will consider the data set obtained from Munnell [24], in which the productivity of public capital in the 48 continental states of the USA is analyzed from 1970 to 1986. According to Greene [11] one has 9 regions, each one consisting of a group of the 48 states. The purpose of this example is to investigate the productivity of public capital according to variables such as the water utility capital and the year of data recording. The response refers to the natural logarithm of the gross state product (LGSP). For each one of the 48 states, one has 17 years of data totalizing a data set with 816 observations. Therefore, one has a longitudinal study with three levels of hierarchy: year, state and region.

We may notice from Figure 3 some variability for the initial gross state product of each region, as well as considering all the regions. Thus, a random intercept for the state giving the region and a random intercept for the region may be considered for modeling the gross state product. For the normal multilevel fitting we consider the model

Level 1 equation:

$$\text{Lgsp}_{ijk} = \gamma_{1j|k} + \gamma_{2j|k} \text{Lwater}_{ijk} + \gamma_{3j|k} \text{Year}_{ijk} + \epsilon_{ijk};$$

Level 2 equations:

$$\gamma_{1j|k} = \delta_{11k} + \xi_{1j|k}; \quad \gamma_{2j|k} = \delta_{21k}; \quad \gamma_{3j|k} = \delta_{31k};$$

Level 3 equations:

$$\delta_{11k} = \beta_{111} + \vartheta_{11k}; \quad \delta_{21k} = \beta_{211}; \quad \delta_{31k} = \beta_{311},$$

for $i = 1, 2, \dots, 17$, $j = 1, 2, \dots, n_k$ and $k = 1, 2, \dots, 9$, where n_k is the number of states in the k th region. Here, Lgsp_{ijk} denotes the natural LGSP for the i th year in the j th state of the k th region, Lwater_{ijk} denotes the natural logarithm of the water utility capital in the i th year in the j th state of the k th region, whereas Year_{ijk} is the i th year in the j th state of the k th region. It is assumed that $\epsilon_{ijk} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, $\xi_{1j|k} \stackrel{\text{iid}}{\sim} N(0, \sigma_\xi^2)$ and $\vartheta_{11k} \stackrel{\text{iid}}{\sim} N(0, \sigma_R^2)$. Also, one has independence among ϵ_{ijk} 's, $\xi_{1j|k}$'s and ϑ_{11k} 's.

In the level 1 equation of the model, we have $\gamma_{1j|k}$ as the intercept for the j th state of the k th region, $\gamma_{2j|k}$ the parameter associated with the logarithm of the water utility capital belonging to the j th state of the k th region, $\gamma_{3j|k}$ the coefficient for the year in the j th state of the k th region and a random error ϵ_{ijk} associated with Lgsp_{ijk} . Regarding the level 2 equation one has that δ_{11k} is the mean of $\gamma_{1j|k}$ in the k th region, δ_{21k} indicates that the effect of water utility capital on the LGSP is the same for all the states in the k th region and δ_{31k} has analogous interpretation of δ_{21k} concerning the year considered. For the level 3 equation one has that β_{111} is the expected value of the random effect δ_{11k} and β_{211} indicates that the effect of water utility capital is the same for all the states and for all the regions. The parameter β_{311} has analogous interpretation of β_{211} regarding the year considered. The model (2) may be expressed, alternatively, as

$$\text{Lgsp}_{ijk} = \beta_{111} + \beta_{211} \text{Lwater}_{ijk} + \beta_{311} \text{Year}_{ijk} + \vartheta_{11k} + \xi_{1j|k} + \epsilon_{ijk}.$$

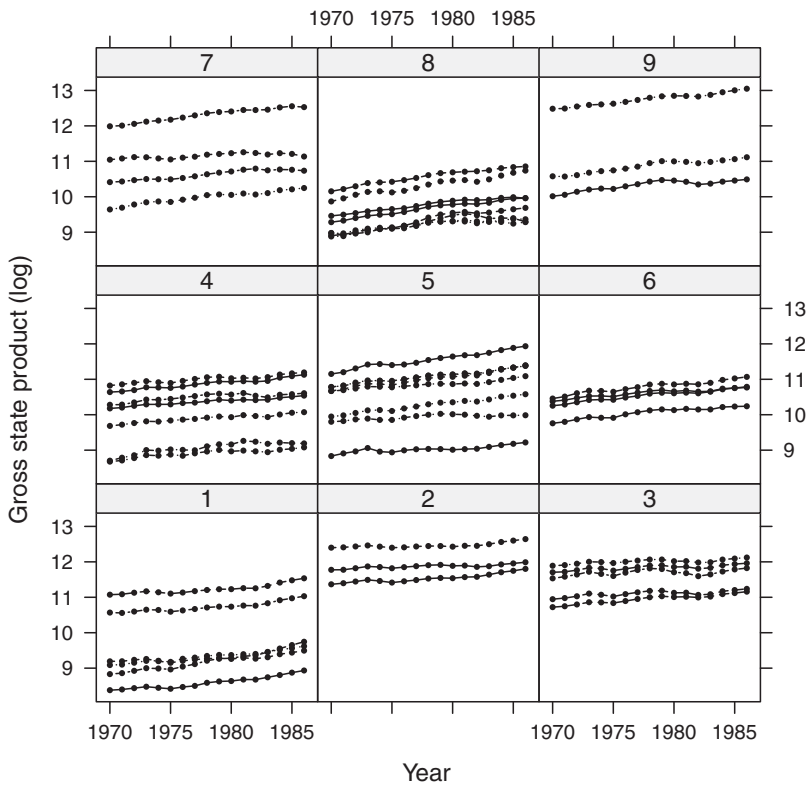


Figure 3. LGSP by region from 1970 to 1986 for the 48 continental states of the USA considered on the Gross State Product data.

Here, β_{111} is the model intercept, β_{211} is the fixed effect of the logarithm of the water utility capital on the expected LGSP and β_{311} is the fixed effect of the year on the expected LGSP. One has also the random effect ϑ_{11k} of the k th region, the random effect $\xi_{1j|k}$ of the j th state in the k th region and the error ϵ_{ijk} associated with the response Lgsp_{ijk} .

Similarly to the first example for goodness-of-fit analysis one may first consider the Mahalanobis distance. So, we may notice from the left graph of Figure 4 region 8 as a possible outlying observation. For this region, the minimum value for the LGSP was 8.8784 whereas the maximum value was 10.8548. From the graph of the normal probability plot with simulated 95% confidence band for the studentized conditional residual we may notice some observations lying outside the superior part of the confidence band. So, a heavy-tailed error model may accommodate better such observations.

3. The elliptical class of distributions

Before defining the elliptical multilevel model, we will introduce the elliptical class of distributions which includes all symmetric continuous distributions. Examples of some of these distributions are the normal, Student- t , slash, power exponential, logistics I and II among others. Their properties may be found, for instance, in Fang *et al.* [6]. We say that

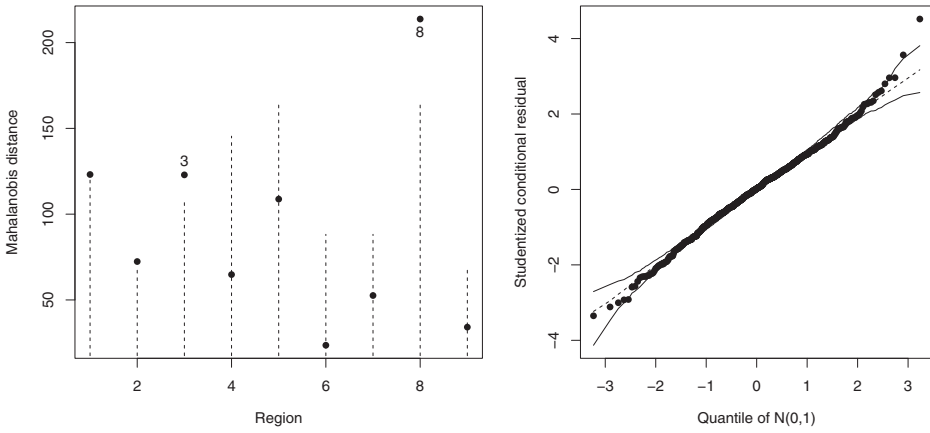


Figure 4. Estimated Mahalanobis distances with their 95% quantiles by region (left) and normal probability plot with simulated 95% confidence band for the studentized conditional residual (right) under the normal multilevel model fitted to the Gross State Product data.

a random vector $\mathbf{Y}_k = (Y_{1k}, Y_{2k}, \dots, Y_{n_k k})^\top$ follows an n_k -dimensional elliptical distribution with mean vector $\boldsymbol{\mu}_k$ and variance-covariance matrix proportional to $\mathbf{V}_k$ assumed here positive-definite, if its density function is given by

$$f(\mathbf{y}_k) = |\mathbf{V}_k|^{-1/2} g(u_k), \quad k = 1, \dots, n,$$

where $u_k = (\mathbf{y}_k - \boldsymbol{\mu}_k)^\top \mathbf{V}_k^{-1} (\mathbf{y}_k - \boldsymbol{\mu}_k)$ is the Mahalanobis distance, the function $g(\cdot) : \mathbb{R} \rightarrow [0, \infty)$ is such that $\int_0^\infty u^{(n_k/2)-1} g(u) du < \infty$, being defined as the density generating function, as seen in Fang *et al.* [6]. When they exist, $E(\mathbf{Y}_k) = \boldsymbol{\mu}_k$ e $\text{Var}(\mathbf{Y}_k) = \alpha_k \mathbf{V}_k$, where α_k is obtained from the characteristic function of $\mathbf{Y}_k$, which takes the form $\varphi_{\mathbf{Y}_k}(\mathbf{t}) = E(e^{i\mathbf{t}^\top \mathbf{Y}_k}) = e^{i\mathbf{t}^\top \boldsymbol{\mu}_k} \phi(\mathbf{t}^\top \mathbf{V}_k \mathbf{t})$, for some function $\phi(\cdot)$. Therefore, we have $\alpha_k = -2\phi'(\mathbf{0})$, where $\phi'(\mathbf{t}) = \partial \phi / \partial \mathbf{t} |_{\mathbf{t}=\mathbf{0}}$. For the Student- t distribution one has $\alpha_k = \eta_k / (\eta_k - 2)$ as seen in Lange *et al.* [19], where $\eta_k > 2$, with η_k being the degrees of freedom. For the power exponential distribution (see Gómez *et al.* [10]) one has $\alpha_k = 2^{(1/\eta_k)} \Gamma\{(n_k + 2)/2\eta_k\} / (n_k \Gamma\{n_k/2\eta_k\})$, where $\Gamma(\cdot)$ is the gamma function and $\eta_k > 0$ is the shape parameter, being defined differently for each elliptical distribution. One has, respectively, light- and heavy-tailed distributions for $\eta_k > 1$ and $0 < \eta_k < 1$, and the normal distribution is recovered for $\eta_k = 1$. A discussion on singular elliptical distributions, that is, when $\mathbf{V}_k$ is a positive-semi-definite matrix, is presented, for instance, in Dí as-García *et al.* [5].

We will denote $\mathbf{Y}_k \sim \text{El}_{n_k}(\boldsymbol{\mu}_k, \mathbf{V}_k, \eta_k)$ and the simplified notation $\mathbf{Y}_k \sim \text{El}_{n_k}(\boldsymbol{\mu}_k, \mathbf{V}_k)$. In the case of $\mathbf{Y}_k \sim \text{El}_{n_k}(\mathbf{0}, \mathbf{I}_{n_k})$, where $\mathbf{I}_{n_k}$ is the identity matrix of order n_k , we have α_k being the variance of the marginal univariate distributions obtained from $\mathbf{Y}_k$. For the normal distribution one has $\alpha_k = 1$.

4. Model specification

The elliptical multilevel models may be defined from the structure defined for the normal multilevel models. Therefore, considering the matrix notation of the multilevel model with three levels, given in the [appendix](#), one has $\mathbf{Y}_k = \mathbf{X}_k \boldsymbol{\beta} + \mathbf{Z}_k \mathbf{b}_k + \boldsymbol{\epsilon}_k$, for $k =$

$1, 2, \dots, K$, and two approaches may be considered. The first would be the use of the hierarchical formulation $\mathbf{Y}_k | \mathbf{b}_k \stackrel{\text{ind}}{\sim} \text{El}_{n_k}(\mathbf{X}_k \boldsymbol{\beta} + \mathbf{Z}_k \mathbf{b}_k, \mathbf{R}_k)$ and $\mathbf{b}_k \stackrel{\text{ind}}{\sim} \text{El}_{z_{1k}}(\mathbf{0}, \boldsymbol{\Psi}_k)$, where $z_{1k} = \sum_{p=1}^P Q_p + Pm_k$. The difficulty of using this approach lies in the fact that the joint distribution of $(\mathbf{Y}_k^\top, \mathbf{b}_k^\top)^\top$ is not necessarily elliptical (except for the normal case), therefore the marginal distribution of $\mathbf{Y}_k$ may be analytically hard to be obtained. However, for the particular case of the multivariate Student- t distribution (see Pinheiro *et al.* [27]), by adding the random variable τ_k in the hierarchical formulation

$$\begin{aligned} \mathbf{Y}_k | (\mathbf{b}_k^\top, \tau_k)^\top &\stackrel{\text{ind}}{\sim} N_{n_k} \left(\mathbf{X}_k \boldsymbol{\beta} + \mathbf{Z}_k \mathbf{b}_k, \frac{1}{\tau_k} \mathbf{R}_k \right); \\ \mathbf{b}_k | \tau_k &\stackrel{\text{ind}}{\sim} N_{z_{1k}} \left(\mathbf{0}, \frac{1}{\tau_k} \boldsymbol{\Psi}_k \right); \\ \tau_k &\stackrel{\text{ind}}{\sim} \text{Gamma} \left(\frac{\eta_k}{2}, \frac{\eta_k}{2} \right), \end{aligned}$$

one has that $\mathbf{Y}_k \stackrel{\text{ind}}{\sim} t_{n_k}(\mathbf{X}_k \boldsymbol{\beta}, \mathbf{Z}_k \boldsymbol{\Psi}_k \mathbf{Z}_k^\top + \mathbf{R}_k, \eta_k)$, where $\mathbf{Y}_k \sim t_{n_k}(\boldsymbol{\mu}_k, \mathbf{V}_k, \eta_k)$ denotes a random vector $\mathbf{Y}_k$ following an n_k -dimensional multivariate Student- t distribution, having mean vector $\boldsymbol{\mu}_k$ and variance-covariance matrix proportional to $\mathbf{V}_k$, with η_k being the respective degrees of freedom.

In case of the multivariate Student- t distribution, the hierarchical approach mentioned above does not imply much difficulties in use. However, for other distributions belonging to the elliptical class, obtaining the marginal distribution of $\mathbf{Y}_k$ is usually more complex, being necessary the use of numerical integration. Thus, another approach may be adopted, by considering

$$\begin{bmatrix} \mathbf{Y}_k \\ \mathbf{b}_k \end{bmatrix} \stackrel{\text{ind}}{\sim} \text{El}_{z_{2k}} \left(\begin{bmatrix} \mathbf{X}_k \boldsymbol{\beta} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \mathbf{Z}_k \boldsymbol{\Psi}_k \mathbf{Z}_k^\top + \mathbf{R}_k & \mathbf{Z}_k \boldsymbol{\Psi}_k \\ \boldsymbol{\Psi}_k \mathbf{Z}_k^\top & \boldsymbol{\Psi}_k \end{bmatrix} \right),$$

for $k = 1, 2, \dots, K$, where $z_{2k} = n_k + z_{1k}$. In this case, from the elliptical distribution properties (see, for instance, Fang *et al.* [6]) we have

$$\mathbf{Y}_k \stackrel{\text{ind}}{\sim} \text{El}_{n_k}(\mathbf{X}_k \boldsymbol{\beta}, \mathbf{Z}_k \boldsymbol{\Psi}_k \mathbf{Z}_k^\top + \mathbf{R}_k). \quad (3)$$

Considering the K units belonging to level 3, we may assume

$$\begin{bmatrix} \mathbf{Y} \\ \mathbf{b} \end{bmatrix} \sim \text{El}_{z_3} \left(\begin{bmatrix} \mathbf{X} \boldsymbol{\beta} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \mathbf{Z} \boldsymbol{\Psi} \mathbf{Z}^\top + \mathbf{R} & \mathbf{Z} \boldsymbol{\Psi} \\ \boldsymbol{\Psi} \mathbf{Z}^\top & \boldsymbol{\Psi} \end{bmatrix} \right),$$

where $z_3 = \sum_{k=1}^K z_{2k}$. Therefore, we have $\mathbf{Y} \sim \text{El}_n(\mathbf{X} \boldsymbol{\beta}, \mathbf{Z} \boldsymbol{\Psi} \mathbf{Z}^\top + \mathbf{R})$, where $n = \sum_{k=1}^K n_k$. Here, $\mathbf{X}$, $\boldsymbol{\beta}$ and $\mathbf{Z}$ are defined as in the [appendix](#), $\boldsymbol{\Psi} = \text{diag}\{\boldsymbol{\Psi}_1, \boldsymbol{\Psi}_2, \dots, \boldsymbol{\Psi}_K\}$ and $\mathbf{R} = \text{diag}\{\mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_K\}$.

4.1. Maximum likelihood estimation

According to the model defined in Equation (3), for $k = 1, 2, \dots, K$, we have the probability density function of $\mathbf{Y}_k$ given by $f(\mathbf{y}_k; \boldsymbol{\beta}, \boldsymbol{\tau}) = |\mathbf{V}_k|^{-(1/2)} g(u_k)$, where $u_k = (\mathbf{y}_k -$

$\mathbf{X}_k \boldsymbol{\beta})^\top \mathbf{V}_k^{-1} (\mathbf{y}_k - \mathbf{X}_k \boldsymbol{\beta})$ and $\mathbf{V}_k = \mathbf{Z}_k \boldsymbol{\Psi}_k \mathbf{Z}_k^\top + \mathbf{R}_k$. Define $\boldsymbol{\tau} = (\tau_1, \tau_2, \dots, \tau_D)^\top$ the vector of distinct parameters in the matrices $\mathbf{V}_k$, for $k = 1, 2, \dots, K$. Therefore, the logarithm of the likelihood function yields

$$l(\boldsymbol{\beta}, \boldsymbol{\tau}) = -\frac{1}{2} \sum_{k=1}^K \log |\mathbf{V}_k| + \sum_{k=1}^K \log \{g(u_k)\}. \quad (4)$$

The partial derivatives of Equation (4) with respect to $\boldsymbol{\beta}$ and τ_d are, respectively, given by

$$\begin{aligned} \mathbf{S}_\beta &= \sum_{k=1}^K v(u_k) \mathbf{X}_k^\top \mathbf{V}_k^{-1} (\mathbf{y}_k - \mathbf{X}_k \boldsymbol{\beta}) \quad \text{and} \\ \mathbf{S}_{\tau_d} &= -\frac{1}{2} \sum_{k=1}^K \left[\text{trace} \left(\mathbf{V}_k^{-1} \frac{\partial \mathbf{V}_k}{\partial \tau_d} \right) - v(u_k) (\mathbf{y}_k - \mathbf{X}_k \boldsymbol{\beta})^\top \mathbf{V}_k^{-1} \frac{\partial \mathbf{V}_k}{\partial \tau_d} \mathbf{V}_k^{-1} (\mathbf{y}_k - \mathbf{X}_k \boldsymbol{\beta}) \right], \end{aligned}$$

for $d = 1, 2, \dots, D$, where $v(u_k) = -2\omega(u_k)$, $\omega(u_k) = g'(u_k)/g(u_k)$ and $g'(u_k) = dg(u_k)/du_k$. By making $\mathbf{S}_\beta = \mathbf{0}$ we obtain the following iterative process for the parameter estimation:

$$\boldsymbol{\beta}^{(r+1)} = \left[\sum_{k=1}^K v(u_k)^{(r)} \mathbf{X}_k^\top \mathbf{V}_k^{-1(r)} \mathbf{X}_k \right]^{-1} \left[\sum_{k=1}^K v(u_k)^{(r)} \mathbf{X}_k^\top \mathbf{V}_k^{-1(r)} \mathbf{y}_k \right] \quad \text{and} \quad (5)$$

$$\boldsymbol{\tau}^{(r+1)} = \arg \max_{\boldsymbol{\tau}} [l(\boldsymbol{\beta}^{(r+1)}, \boldsymbol{\tau})], \quad (6)$$

for $r = 0, 1, 2, \dots$. Starting values should be used in the estimation process. Note that this process is equivalent to the one used for normal multilevel models with $\sqrt{v(u_k)} \mathbf{Y}_k \sim N_{n_k}(\sqrt{v(u_k)} \mathbf{X}_k \boldsymbol{\beta}, \mathbf{V}_k)$. Thus, given $v(u_k)$, computational routines already implemented for normal multilevel models may be used for obtaining updates for $\boldsymbol{\beta}$ and $\boldsymbol{\tau}$ from Equations (5) and (6). The quantity $v(u_k)$ appearing in Equation (5) may be seen as a weight assigned for each cluster in the 3rd hierarchy level from the estimation process. Since the function $g(u_k)$ is in most of the cases a decreasing positive function, we have in general $v(u_k) > 0$. For the Student- t distribution one has $v(u_k) = (\eta_k + n_k)/(\eta_k + u_k)$ and for the power exponential distribution $v(u_k) = \eta_k u_k^{\eta_k - 1}$, as seen in Osorio *et al.* [25]. In particular, $v(u_k)$ is inversely proportional to the Mahalanobis distance for the Student- t and heavy-tailed power exponential distributions. Therefore, larger values of u_k imply smaller values of $v(u_k)$ and consequently smaller weights will be given for outlying clusters in the parameter estimation process.

The Fisher information matrices for $\boldsymbol{\beta}$ and $\boldsymbol{\tau}$ are given, respectively, by $\mathbf{K}_{\beta\beta} = \sum_{k=1}^K (4d_{gk}/n_k) \mathbf{X}_k^\top \mathbf{V}_k^{-1} \mathbf{X}_k$ and $\mathbf{K}_{\tau\tau} = \{\kappa_{\tau_l \tau_z}\}$, where $d_{gk} = E\{\omega(u_k)^2 u_k\}$ and $\kappa_{\tau_l \tau_z}$ is the element belonging to the l th row and z th column of the matrix $\mathbf{K}_{\tau\tau}$, given by

$$\kappa_{\tau_l \tau_z} = \sum_{k=1}^K [-c_{klz}/4 + \{f_{gk}/n_k(n_k + 2)\} \{c_{klz} + 2\text{trace}(\mathbf{B}_k)\}],$$

where $c_{klz} = \text{trace}(\mathbf{V}_k^{-1} \partial \mathbf{V}_k / \partial \tau_l) \text{trace}(\mathbf{V}_k^{-1} \partial \mathbf{V}_k / \partial \tau_z)$, $f_{gk} = E\{\omega(u_k)^2 u_k^2\}$ for $l, z = 1, 2, \dots, D$ and $\mathbf{B}_k = \mathbf{V}_k^{-1} (\partial \mathbf{V}_k / \partial \tau_l) \mathbf{V}_k^{-1} (\partial \mathbf{V}_k / \partial \tau_z)$. Considering the Student- t distribution with η_k degrees of freedom we have $d_{gk} = (n_k/4) \{(\eta_k + n_k)/(\eta_k + n_k + 2)\}$ and

$f_{gk} = \{n_k(n_k + 2)/4\}\{(\eta_k + n_k)/(\eta_k + n_k + 2)\}$. For the power exponential distribution with shape parameter η_k we have $d_{gk} = (\eta_k^2/2^{(1/\eta_k)})\Gamma\{(n_k - 2)/2\eta_k + 2\}/\Gamma(n_k/2\eta_k)$ and $f_{gk} = n_k(n_k + 2\eta_k)/4$ (see, for instance, Osorio *et al.* [25]).

4.2. Random effect prediction

Similar to the normal case, one may consider for the prediction of the random effects the distribution of b_k given the observed data $\mathbf{y}_k$. According to Fang *et al.* [6], we have that $b_k | \mathbf{Y}_k = \mathbf{y}_k$ follows an elliptical distribution. When the vector $\boldsymbol{\tau}$ is known, we may use the empirical Bayes predictor $\hat{b}_k = \boldsymbol{\Psi}_k \mathbf{Z}_k^\top \mathbf{V}_k^{-1} (\mathbf{y}_k - \mathbf{X}_k \hat{\boldsymbol{\beta}})$, for $k = 1, 2, \dots, K$. In case of $\boldsymbol{\tau}$ unknown, its maximum likelihood estimate will be used in the matrices $\boldsymbol{\Psi}_k$ and $\mathbf{V}_k$. Considering the K level 3 units, we have $\hat{\mathbf{b}} = \boldsymbol{\Psi} \mathbf{Z}^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}})$, where $\mathbf{y} = (\mathbf{y}_1^\top, \mathbf{y}_2^\top, \dots, \mathbf{y}_K^\top)^\top$.

5. Residual analysis

In order to detect the presence of outlying observations as well as departures from the model assumptions, one may consider the marginal and conditional residuals, namely $\hat{\mathbf{r}} = \mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\epsilon}} = \mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{Z} \hat{\mathbf{b}}$, respectively. Here, $\hat{\boldsymbol{\beta}}$ is the maximum likelihood estimate of $\boldsymbol{\beta}$ and $\hat{\mathbf{b}}$ is the linear predictor of $\mathbf{b}$. These residuals may be applied for studying the behavior of the errors $\mathbf{r} = \mathbf{Y} - \mathbf{X} \boldsymbol{\beta} = \mathbf{Z} \mathbf{b} + \boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon} = \mathbf{Y} - \mathbf{X} \boldsymbol{\beta} - \mathbf{Z} \mathbf{b}$, respectively. One may also use the linear predictor $\hat{\mathbf{b}} = \boldsymbol{\Psi} \mathbf{Z}^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}})$ for assessing the behavior of the errors in $\mathbf{b}$.

As in the normal case, index plots of the marginal and conditional residuals may reveal the presence of outlying observations in the sense of their respective errors. However, such analyzes should be performed by considering the estimates of the weights $v(u_k)$'s for each cluster, since outlying observations should receive small weights compared to the others from the iterative process, and thus their influence on the parameter estimates should be reduced. Also, the index plot of the the linear predictor may reveal outlying clusters with respect to the random effects.

Because the above residuals may present different variances, approximate standardized forms will be derived for them. Studentized marginal and conditional forms are, respectively, given by $\hat{r}_{ijk}^* = \hat{r}_{ijk} / \widehat{\text{se}}(\hat{r}_{ijk})$ and $\hat{\epsilon}_{ijk}^* = \hat{\epsilon}_{ijk} / \widehat{\text{se}}(\hat{\epsilon}_{ijk})$, whereas studentized linear predictors yield $\hat{b}_{ijk}^* = \hat{b}_{ijk} / \widehat{\text{se}}(\hat{b}_{ijk})$, where $\widehat{\text{se}}(\hat{r}_{ijk})$, $\widehat{\text{se}}(\hat{\epsilon}_{ijk})$ and $\widehat{\text{se}}(\hat{b}_{ijk})$ denote the estimated standard errors of $\hat{r}_{ijk}$, $\hat{\epsilon}_{ijk}$ and $\hat{b}_{ijk}$, respectively.

Setting $v(u_k) = v_k$ and by considering v_k and $\boldsymbol{\tau}$ fixed, one may write the maximum likelihood estimate of $\boldsymbol{\beta}$, at the convergence of the iterative process, as $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{M} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{M} \mathbf{Y}$, where $\mathbf{M} = \text{diag}\{v_1 \mathbf{V}_1^{-1}, v_2 \mathbf{V}_2^{-1}, \dots, v_K \mathbf{V}_K^{-1}\}$. Then, the marginal and conditional residuals may be expressed as $\hat{\mathbf{r}} = \mathbf{M}^{-1} \mathbf{Q} \mathbf{Y}$ and $\hat{\boldsymbol{\epsilon}} = \mathbf{A} (\mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}})$, respectively, where $\mathbf{Q} = \mathbf{M} - \mathbf{M} \mathbf{X} (\mathbf{X}^\top \mathbf{M} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{M}$, $\mathbf{I}_n$ is the identity matrix of order n and $\mathbf{A} = \mathbf{I}_n - \mathbf{Z} \boldsymbol{\Psi} \mathbf{Z}^\top \mathbf{V}^{-1}$. The variance-covariance matrices for the marginal and conditional residuals are given, when v_k and $\boldsymbol{\tau}$ are known, by

$$\text{Var}(\hat{\mathbf{r}}) = \mathbf{M}^{-1} \mathbf{Q} \mathbf{V}^* \mathbf{Q} \mathbf{M}^{-1}$$

and

$$\text{Var}(\hat{\epsilon}) = \mathbf{A}\mathbf{M}^{-1}\mathbf{Q}\mathbf{V}^*\mathbf{Q}\mathbf{M}^{-1}\mathbf{A}^\top,$$

respectively, whereas for the linear predictors the variance–covariance matrix takes the form

$$\text{Var}(\hat{\mathbf{b}}) = \boldsymbol{\Psi}\mathbf{Z}^\top\mathbf{V}^{-1}\mathbf{M}^{-1}\mathbf{Q}\mathbf{V}^*\mathbf{Q}\mathbf{M}^{-1}\mathbf{V}^{-1}\mathbf{Z}\boldsymbol{\Psi},$$

where $\mathbf{V}^* = \text{diag}\{\alpha_1\mathbf{V}_1, \alpha_2\mathbf{V}_2, \dots, \alpha_K\mathbf{V}_K\}$. Thus, when v_k and $\boldsymbol{\tau}$ are unknown, we may derive the studentized residuals from the diagonal elements of the respective variance–covariance estimated matrices.

For assessing departures from the postulated error distributions in multilevel elliptical models, normal probability plots with simulated confidence bands for the empirical distributions of $\hat{r}_{ijk}^*$, $\hat{\epsilon}_{ijk}^*$ and $\hat{b}_{ijk}^*$ may be performed. Such graphs may reveal, for instance, if it is reasonable to assume light- or heavy-tailed error distributions as well as presence of outlying observations in the 1st hierarchy level.

Alternatively, for assessing the presence of outlying clusters in the 3rd hierarchy level one may consider the Mahalanobis distance $u_k = (\mathbf{Y}_k - \boldsymbol{\mu}_k)^\top \mathbf{V}_k^{-1} (\mathbf{Y}_k - \boldsymbol{\mu}_k)$ under the proposed error model. In particular, under the Student- t multilevel model with η_k degrees of freedom, $u_k^t = u_k/n_k \sim F(n_k, \eta_k)$ (see, for instance, Lange *et al.* [19]), whereas under the power exponential multilevel model with shape parameter η_k , $u_k^p = u_k^{\eta_k} \sim \text{Gamma}(n_k/2\eta_k, 1/2)$ (see, for instance, Gómes *et al.* [10]). It may be informative to perform the plot of such distances against the v_k estimates.

6. Simulation study

This section presents a small simulation study to assess some statistical properties of the maximum likelihood estimates under elliptical multilevel models. For the purpose of illustration, we will consider the model

Level 1 equation:

$$y_{ijk} = \gamma_{1j|k} + \gamma_{2j|k}x_{ijk} + \epsilon_{ijk};$$

Level 2 equations:

$$\gamma_{1j|k} = \delta_{11k} + \xi_{1j|k}; \quad \gamma_{2j|k} = \delta_{21k};$$

Level 3 equations:

$$\delta_{11k} = \beta_{111} + \vartheta_{11k}; \quad \delta_{21k} = \beta_{211},$$

for $i = 1, 2, \dots, 5$, $j = 1, 2, \dots, 5$ and $k = 1, 2, \dots, n$. Here, x_{ijk} were generated from a uniform distribution in the interval $[0, 1]$, $\epsilon_{ijk} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, $\xi_{1j|k} \stackrel{\text{iid}}{\sim} N(0, \sigma_\xi^2)$ and $\vartheta_{11k} \stackrel{\text{iid}}{\sim} N(0, \sigma_\vartheta^2)$, with independence among ϵ_{ijk} 's, $\xi_{1j|k}$'s and ϑ_{11k} 's. The true parameter values are $\sigma^2 = \sigma_\xi^2 = \sigma_\vartheta^2 = \beta_{111} = \beta_{211} = 1$. Therefore, one has the intraclass correlation coefficients (ICCs) $\text{ICC}_1 = \sigma_\vartheta^2 / (\sigma_\vartheta^2 + \sigma_\xi^2 + \sigma^2) = \frac{1}{3}$ and $\text{ICC}_2 = (\sigma_\vartheta^2 + \sigma_\xi^2) / (\sigma_\vartheta^2 + \sigma_\xi^2 + \sigma^2) = \frac{2}{3}$. To consider different simulation scenarios, different elliptical distributions were used [i.e. multivariate normal, Student- t ($\eta = 5, 10$ and 20) and power-exponential ($\eta =$

Table 1. Relative bias (in absolute values) of the maximum likelihood estimates from a simulation study of elliptical multilevel models.

Fitted model	<i>n</i>											
	$\hat{\beta}_{111}$			$\hat{\beta}_{211}$			$\widehat{ICC}_1$			$\widehat{ICC}_2$		
	20	30	50	20	30	50	20	30	50	20	30	50
Normal	0.07	0.05	0.03	0.25	0.06	0.02	0.29	0.17	0.10	0.24	0.13	0.08
Student- <i>t</i> (5)	0.00	0.00	0.00	0.02	0.00	0.00	0.13	0.07	0.04	0.06	0.03	0.02
Student- <i>t</i> (10)	0.00	0.01	0.00	0.03	0.01	0.00	0.13	0.07	0.04	0.06	0.03	0.02
Student- <i>t</i> (20)	0.00	0.01	0.01	0.04	0.01	0.00	0.14	0.08	0.04	0.05	0.02	0.01
Power exp. (0.3)	0.01	0.01	0.01	0.05	0.01	0.00	0.15	0.08	0.04	0.08	0.04	0.02
Power exp. (0.5)	0.02	0.02	0.01	0.08	0.02	0.01	0.17	0.10	0.05	0.11	0.06	0.03
Power exp. (0.9)	0.05	0.04	0.02	0.20	0.05	0.02	0.26	0.15	0.09	0.20	0.11	0.07

0.3, 0.5 and 0.9)], and the sample size is modified, that is, by considering $n = 20, 30$ and 50. This process was replicated $N = 5,000$ times, keeping the value of the variable x fixed. In order to obtain at least an outlying observation, under each replication, the last simulated response value was modified as $y_{55n}^* = y_{55n} + 10\sigma_N$, where σ_N denotes the standard deviation of the N th simulated response vector. Summary measures of the relative biases of $\hat{\beta}_{111}$, $\hat{\beta}_{211}$, $\widehat{ICC}_1$ and $\widehat{ICC}_2$ were calculated. In addition, a third summary measure, the statistic of the Kolmogorov–Smirnov goodness-of-fit test (see Conover [1]), for assessing the normality of the sample estimates, was evaluated.

The values of the KS statistic for $\hat{\beta}_{111}$ and $\hat{\beta}_{211}$ (omitted here) became near to zero under all parameter settings and sample sizes, which indicate that such estimates follow approximately normal distributions. According to the remaining results presented in Table 1, we may notice that, in general, the relative biases (in absolute values) of $\hat{\beta}_{111}$, $\hat{\beta}_{211}$, $\widehat{ICC}_1$ and $\widehat{ICC}_2$ decreases as the sample size increases. In addition, under Student-*t* and power exponential distributions, the relative biases of $\widehat{ICC}_1$ and $\widehat{ICC}_2$ decreases as the kurtosis coefficient decreases.

7. Applications

7.1. The Rat Pup study

For this application we will consider the same model given in Equation (1), but with the following joint multivariate distribution:

$$\begin{bmatrix} \mathbf{Y}_k \\ b_{1k} \end{bmatrix} \stackrel{\text{ind}}{\sim} \text{El}_{k_1} \left(\begin{bmatrix} \mathbf{X}_k \boldsymbol{\beta} \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbf{Z}_k \boldsymbol{\Psi}_k \mathbf{Z}_k^\top + \mathbf{R}_k & \mathbf{Z}_k \boldsymbol{\Psi}_k \\ \boldsymbol{\Psi}_k \mathbf{Z}_k^\top & \boldsymbol{\Psi}_k \end{bmatrix} \right),$$

for $k = 1, 2, \dots, 27$, where $k_1 = n_k + 1$, $\mathbf{Y}_k = (\text{Weight}_{1k}, \dots, \text{Weight}_{n_k k})^\top$, $\mathbf{X}_k$ is an $n_k \times 5$ matrix in which the first column is formed by ones and the remaining columns contain the values of the explanatory variables Treat1, Treat2, Litsize and Sex, respectively, for the rat pup in the k th litter, whereas $\mathbf{Z}_k$ is an $n_k \times 1$ vector of ones and $\boldsymbol{\beta} = (\beta_{11}, \beta_{12}, \beta_{13}, \beta_{14}, \beta_{21})^\top$. In addition, $\boldsymbol{\Psi}_k = \sigma_L^2$ and $\mathbf{R}_k = \sigma_{CT}^2 \mathbf{I}_{n_k}$, if the k th litter has received the control treatment or $\mathbf{R}_k = \sigma_{HL}^2 \mathbf{I}_{n_k}$ if the k th litter has received the low or high dosages of the experimental compound, where $\mathbf{I}_{n_k}$ denotes the identity matrix of order n_k .

Table 2. Parameter estimates and their approximate standard errors under the normal, Student-*t* and power exponential multilevel models fitted to the Rat Pup data.

Effect	Normal	Student- <i>t</i>	Power Exp.
β_{11} Intercept	8.3280 (0.2593)	8.1613 (0.2508)	8.2365 (0.1914)
β_{12} Treat1	−0.8648 (0.1715)	−0.9050 (0.1652)	−0.8496 (0.1524)
β_{13} Treat2	−0.4344 (0.1424)	−0.4444 (0.1359)	−0.4316 (0.1379)
β_{14} Litsize	−0.1307 (0.0175)	−0.1200 (0.0171)	−0.1258 (0.0130)
β_{21} Female	−0.3419 (0.0422)	−0.3004 (0.0375)	−0.3154 (0.0408)
σ^2_L	0.0832 (0.0265)	0.0732 (0.0256)	0.0068 (0.0022)
σ^2_{CT}	0.2639 (0.0339)	0.1166 (0.0289)	0.0157 (0.0026)
σ^2_{HL}	0.0913 (0.0098)	0.0782 (0.0157)	0.0073 (0.0010)
AIC	347.7734	292.9033	334.4138

Table 2 presents the maximum likelihood estimates and their respective approximate standard errors under normal, Student-*t* and power exponential models. The degrees of freedom parameter under the Student-*t* model was selected by the AIC (Akaike information) criterion as $\hat{\eta} = 5$, whereas under the power exponential model the shape parameter selected by the same criterion equals $\hat{\eta} = 0.6$, indicating heavier tails than the normal ones. Comparing the estimates we notice the same inference under the three fitted models. The rat pup groups that received high (Treat1) or low (Treat2) doses present, respectively, the smaller weights, the female group has smaller mean weight than the male group and the mean weight of the rat pups decreases as the litter size increases. However, the approximate standard errors under Student-*t* and power exponential models are systematically smaller than the ones under the normal model, and the Student-*t* model presents the smallest AIC. The application of the likelihood ratio test for assessing $H_0 : \sigma^2_L = 0$ against $H_1 : \sigma^2_L > 0$ (see, for instance, Silvapulle and Silvapulle [31]), which follows for large K a mixture of central chi-squared distributions $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$, where χ^2_0 denotes the degenerate distribution at the origin, equals $\zeta_{LR} = 130.75$, $\zeta_{LR} = 104.45$ and $\zeta_{LR} = 100.46$, respectively, under the the normal, Student-*t* and power exponential models. For these values we found $p < 0.01$, confirming the random effect of litter.

Table 3 presents the estimates of the ICC obtained from the normal, Student-*t* and power exponential fitted models. The quantity ICC_{CT} denotes the intraclass correlation among the rat pups in the litters that received the control treatment, whereas ICC_{HL} denotes the intraclass correlation among the rat pups in the litters that received low or high doses. We notice that the heavy-tailed multilevel models seem to capture better the ICC's in the control group than the normal multilevel model.

Marginally, we notice from Figure 5 that under a 5% significance level litter 6 is pointed out as outlier under the Student-*t* model, whereas litters 5, 6 and 18 are pointed out under the power exponential model.

Table 3. Estimated ICC's under the normal, Student-*t* and power exponential multilevel models fitted to the Rat Pup data.

ICC	Normal	Student- <i>t</i>	Power Exp.
ICC_{CT}	0.2397	0.3858	0.3004
ICC_{HL}	0.4767	0.4837	0.4799

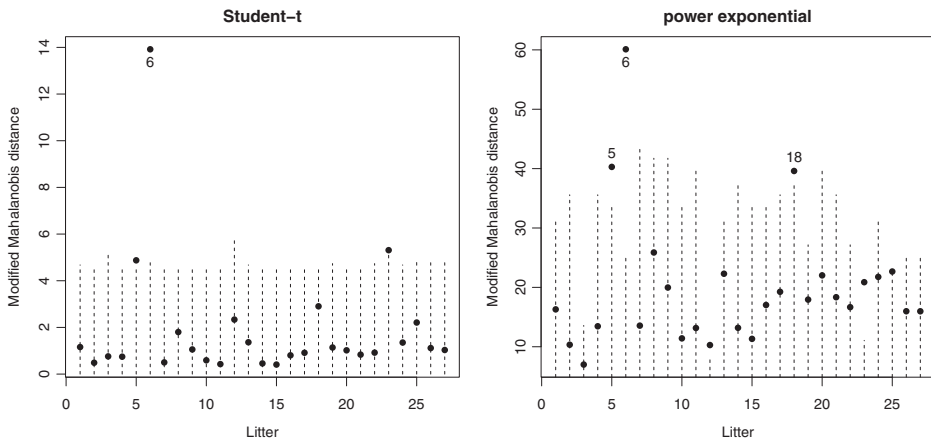


Figure 5. Index plots (by litter) of the estimated modified Mahalanobis distance under the Student- t and power exponential multilevel models fitted to the Rat Pup data.

Table 4. Estimated weights according to the litters under the Student- t and power exponential multilevel models fitted to the Rat Pup data.

Litter	1	2	3	4	5	6	7	8	9
Student- t	2.7603	3.4629	1.5628	3.3071	1.8228	0.7400	4.1809	3.2346	3.6331
Power Exp.	0.1965	0.2357	0.2755	0.2122	0.1368	0.1166	0.2115	0.1633	0.1811
Litter	10	11	12	13	14	15	16	17	18
Student- t	3.2193	3.8690	0.9542	2.6715	3.6654	3.3285	3.1012	3.2126	2.5302
Power Exp.	0.2266	0.2140	0.2362	0.1733	0.2139	0.2272	0.1930	0.1838	0.1377
Litter	19	20	21	22	23	24	25	26	27
Student- t	2.4434	3.4891	3.2568	2.5338	0.7760	2.6768	1.8032	2.2893	2.3208
Power Exp.	0.1891	0.1742	0.1874	0.1947	0.1780	0.1750	0.1722	0.1980	0.1981

Table 4 presents the estimated litter weights from the Student- t and power exponential fitted models and we may notice that the smaller weights (indicated in bold) were assigned to the litters with the largest modified Mahalanobis distances. Such results confirm the downweighting property of the maximum likelihood estimation under heavy-tailed multilevel error models in the sense of the Mahalanobis distance.

From Figure 6 we notice no unusual departure under the Student- t model, all points lie within the simulated confidence band. However, under the power exponential model, one has similar behavior to the ones observed for the studentized marginal (omitted here) and conditional residuals under the normal model, with some points lying outside the simulated confidence band. Thus, these graphs confirm the superiority of the Student- t multilevel model with 5 degrees of freedom to fit the Rat Pup data.

7.2. Gross state product data

Similarly to the first application we will consider for analyzing the Gross State Product data the same model given in Equation (2), but with the following joint multivariate

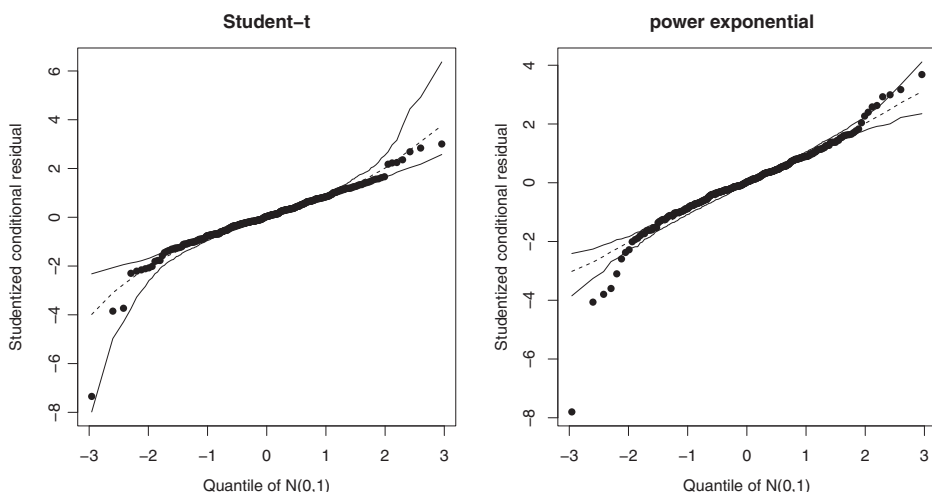


Figure 6. Normal probability plots with 95% simulated confidence bands for the studentized conditional residual under the Student- t and power exponential multilevel models fitted to the Rat Pup data.

distribution:

$$\begin{bmatrix} \mathbf{Y}_k \\ \mathbf{b}_k \end{bmatrix} \underset{\text{ind}}{\sim} \text{El}_{k_2} \left(\begin{bmatrix} \mathbf{X}_k \boldsymbol{\beta} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \mathbf{Z}_k \boldsymbol{\Psi}_k \mathbf{Z}_k^\top + \mathbf{R}_k & \mathbf{Z}_k \boldsymbol{\Psi}_k \\ \boldsymbol{\Psi}_k \mathbf{Z}_k^\top & \boldsymbol{\Psi}_k \end{bmatrix} \right),$$

for $k = 1, 2, \dots, 9$, where $k_2 = 17n_k + 1 + n_k$. Here one has $\mathbf{Y}_k = (\mathbf{Y}_{1k}^\top, \mathbf{Y}_{2k}^\top, \dots, \mathbf{Y}_{n_k k}^\top)^\top$, $\mathbf{Y}_{jk} = (\text{Lgsp}_{1jk}, \dots, \text{Lgsp}_{17jk})^\top$, for $j = 1, 2, \dots, n_k$, $\mathbf{X}_k$ is a $17n_k \times 3$ matrix in which the first column is formed by ones and the remaining columns contain the values of the explanatory variables *Lwater* and *Year* for the k th region, $\mathbf{Z}_k$ is a $17n_k \times (1 + n_k)$ matrix with the first column formed by $17n_k$ ones, the other columns formed by 17 ones and $17(n_k - 1)$ zeros, $\boldsymbol{\beta} = (\beta_{111}, \beta_{211}, \beta_{311})^\top$ and $\mathbf{b}_k = (\vartheta_{11k}, \xi_{11|k}, \xi_{12|k}, \dots, \xi_{1n_k|k})^\top$. In addition, $\boldsymbol{\Psi}_k = \text{diag}\{\sigma_R^2, \sigma_S^2 \mathbf{I}_{n_k}\}$ and $\mathbf{R}_k = \sigma^2 \mathbf{I}_{17n_k}$, where $\mathbf{I}_{n_k}$ is the identity matrix of order n_k .

The maximum likelihood estimates of the parameters with their respective approximate standard errors are given in Table 5 under the normal, Student- t and power exponential models. The degrees of freedom selected by the AIC criterion for the Student- t model was $\hat{\eta} = 12$, whereas the shape parameter selected by the same criterion for the power exponential model equals $\hat{\eta} = 0.9$, both indicating for heavy-tailed error models. Although the parameter estimates present some differences among the three fitted models, the inference is similar. The LGSP increases as the logarithm of water utility increases across the years. Also, in this application, the approximate standard errors are smaller under the heavy-tailed error models and the smallest AIC was obtained under the Student- t model. From Table 6 we may notice that the estimated intraclass correlation among the states is very similar among the three fitted models, however the intraclass correlation among the regions seems to be less evidenced under the Student- t model.

For the assessment of variance components, one may apply the likelihood ratio statistics under some hypothesis (omitted here), where we notice strong evidence for rejecting the absence of state random effect and region random effect. According to Savalli *et al.*

Table 5. Parameter estimates and their approximate standard errors under the normal, Student-*t* and power exponential multilevel models fitted to the Gross State Product data.

Effect	Normal	Student- <i>t</i>	Power Exp.
β_{111} Intercept	-29.6200 (1.9399)	-29.7745 (1.7811)	-29.4278 (1.9279)
β_{211} Lwater	0.2476 (0.0288)	0.2390 (0.0265)	0.2474 (0.0288)
β_{311} Year	0.0194 (0.0011)	0.0195 (0.0010)	0.0193 (0.0011)
σ_R^2	0.2392 (0.1449)	0.1645 (0.1128)	0.1280 (0.0779)
σ_S^2	0.3244 (0.0735)	0.3240 (0.0866)	0.1768 (0.0402)
σ^2	0.0052 (0.0003)	0.0043 (0.0006)	0.0027 (0.0001)
AIC	-1625.0180	-1665.0670	-1633.2650

Table 6. Estimated ICC's under the normal, Student-*t* an power exponential multilevel models fitted to the Gross State Product data.

ICC	Normal	Student- <i>t</i>	Power Exp.
ICC _R	0.4205	0.3338	0.4161
ICC _S	0.9909	0.9914	0.9911

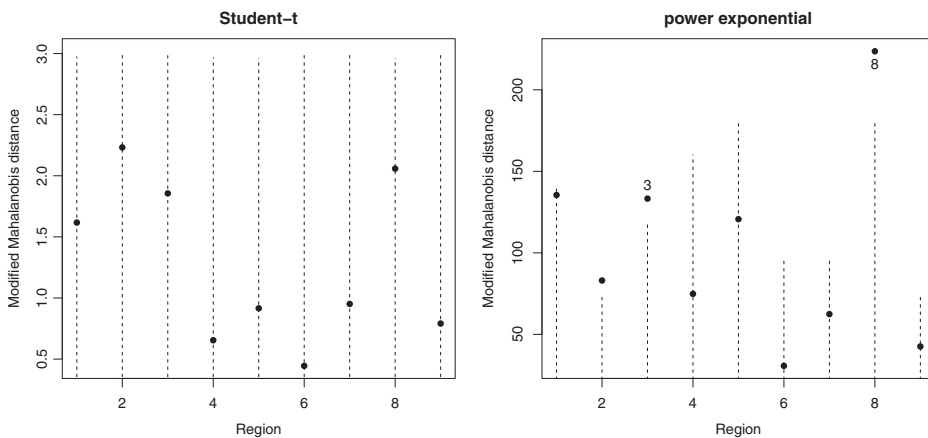


Figure 7. Index plots of the estimated modified Mahalanobis distance according to the region under the Student-*t* and power exponential multilevel models fitted to the Gross State Product data.

[29] and Silvapulle and Silvapulle [31] one has under $H_0 : \sigma_R^2 = \sigma_S^2 = 0$ that the likelihood ratio statistics follows, for large K , a mixture of central chi-squared distributions $\omega_0 \chi_0^2 + \omega_1 \chi_1^2 + \omega_2 \chi_2^2$, where $\omega_0 = (1/2\pi) \cos^{-1}(\rho_{RS})$, $\omega_1 = \frac{1}{2}$ and $\omega_2 = \frac{1}{2} - (1/2\pi) \cos^{-1}(\rho_{RS})$, with ρ_{RS} being the asymptotic correlation between $\hat{\sigma}_R^2$ and $\hat{\sigma}_S^2$ and χ_0^2 denotes the degenerate distribution at the origin. In addition, under $H_0 : \sigma_R^2 = 0$ (with $\sigma_S^2 > 0$) and under $H_0 : \sigma_S^2 = 0$ (with $\sigma_R^2 > 0$), both likelihood ratio statistics follow, for large K , the mixture $\frac{1}{2} \chi_0^2 + \frac{1}{2} \chi_1^2$.

From Figure 7 we notice, marginally, no outstanding region under the Student-*t* model, whereas region 8 is pointed out as outlier under the power exponential model. However, according to Table 7, the smaller weight (indicated in bold) was assigned for the region 8 under the power exponential model, confirming the downweighting property of the maximum likelihood estimation under heavy-tailed multilevel error models in the sense of the Mahalanobis distance.

Table 7. Estimated weights according to the region under the Student-*t* and power exponential multilevel models fitted to the Gross State Product data.

Region	1	2	3	4	5	6	7	8	9
Student- <i>t</i>	8.4909	4.6025	7.1998	10.3099	11.3152	6.4243	6.2404	10.6278	4.8639
Power Exp.	0.5521	0.5791	0.5602	0.5796	0.5517	0.6356	0.6024	0.5244	0.6092

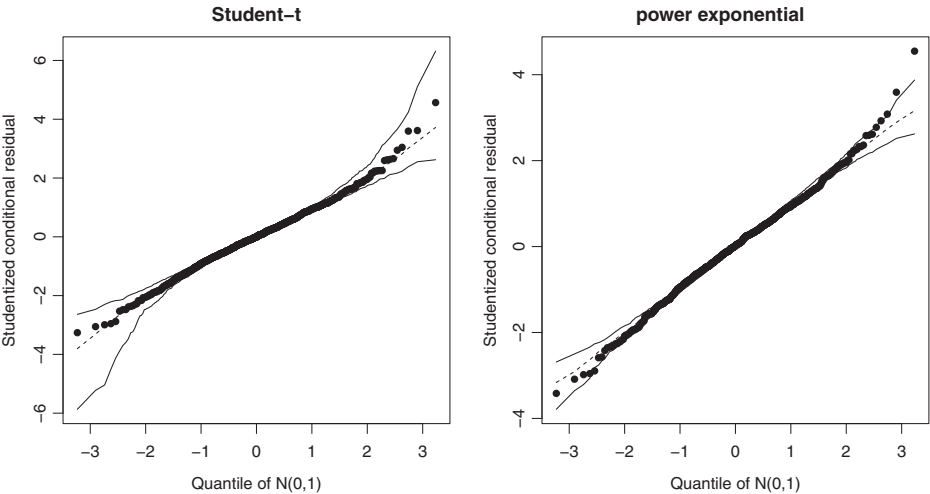


Figure 8. Normal probability plots with 95% simulated confidence bands for the studentized conditional residual under the Student-*t* and power exponential multilevel models fitted to the Gross State Product data.

No unusual departure under the Student-*t* error model for the studentized marginal (omitted here) and conditional residuals is observed in Figure 8, with the majority of points lying within the simulated confidence band for the marginal residuals and all points lying within the simulated confidence band for the conditional residuals. However, under the power exponential model, one has similar behavior of those observed for the studentized marginal (omitted here) and conditional residuals under the normal model, with some points lying outside the simulated confidence band for the marginal and conditional residuals. Thus, these graphs confirm the superiority of the Student-*t* multilevel model with 12 degrees of freedom to fit the Gross State Product data.

8. Concluding remarks

In this paper we extend multilevel models with normal errors to elliptical errors in order to permit distributions with heavier or lighter tails than the normal ones. The proposed elliptical multilevel class has some interesting features, such as (i) closed-form expression for the marginal model, (ii) preservation of the mean and variance-covariance structures of the hierarchical model, (iii) possibility of kurtosis flexibility for the distribution of each response vector, (iv) robust aspects of the maximum likelihood estimates under heavy-tailed errors, and (v) random-effects prediction in a similar way of the normal case. However, it is important to point out from the iterative process (5)–(6) that the downweighting property is made in the *k*th hierarchy level. Thus, in the Rat Pup data study

the downweighting is made in the second hierarchy level (litter), whereas in the Gross State Product data the downweighting is made in the third hierarchy level (region). In general, the goodness-of-fit analysis has been based on the modified Mahalanobis distance in the k th hierarchy level, but there is the possibility of performing a residual analysis in the first level of hierarchy as was made in the two applications.

The codes for the development of the applications in the software R, available in the address <http://www.r-project.org> under the normal, Student- t and power exponential multilevel models, may be available from the authors upon request. The Rat Pup data and the Gross State Product data are available in the addresses <http://www-personal.umich.edu/bwest/chapter4.html> and <http://www.wiley.com/legacy/wileychi/baltagi/supp/PRODUC.prn>, respectively.

Acknowledgments

The authors are grateful to two reviewers for their helpful comments and suggestions.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was partially supported by the Brazilian agencies FACEPE, FAPESP, CAPES and CNPq.

References

- [1] W.J. Conover, *Practical Nonparametric Statistics*, Wiley, New York, 1971.
- [2] F.J.A. Cysneiros and G.A. Paula, *One-sided tests in linear models with multivariate t -distribution*, Comm. Statist. Simulation 33 (2004), pp. 747–771.
- [3] A.P. Dempster, N.M. Laird, and D.B. Rubin, *Maximum likelihood from incomplete data via the EM algorithm*, J. R. Stat. Soc. B 39 (1977), pp. 1–38.
- [4] A.P. Dempster, D.B. Rubin, and R.K. Tsutakawa, *Estimation in covariance components models*, J. Amer. Statist. Assoc. 76 (1981), pp. 341–353.
- [5] J.A. Díaz-García, M. Galea Rojas, and V. Leiva-Sánchez, *Influence diagnostics for elliptical multivariate linear regression models*, Comm. Statist. Theory Methods 32 (2003), pp. 625–641.
- [6] K.T. Fang, S. Kotz, and K.W. Ng, *Symmetric Multivariate and Related Distributions*, Chapman and Hall, New York, 1990.
- [7] C. Fernández and M.F.J. Steel, *Multivariate Student- t regression models: pitfalls and inference*, Biometrika 86 (1999), pp. 153–167.
- [8] M. Galea, G.A. Paula, and H. Bolfarine, *Local influence in elliptical linear regression models*, J. R. Stat. Soc. D 46 (1997), pp. 71–79.
- [9] M. Galea, G.A. Paula, and M. Uribe-Opazo, *On influence diagnostic in univariate elliptical linear regression models*, Statist. Papers 44 (2003), pp. 23–45.
- [10] E. Gómez, M.A. Gómez-Villegas, and J.M. Marín, *A multivariate generalization of the power exponential family of distributions*, Comm. Statist. Theory Methods 27 (1998), pp. 589–600.
- [11] W.H. Greene, *Econometric Analysis*, Prentice Hall, New York, 2012.
- [12] D.A. Harville, *Maximum likelihood approaches to variance component estimation and to related problems*, J. Amer. Statist. Assoc. 72 (1977), pp. 320–338.
- [13] J.A. Hilden-Minton, *Multilevel diagnostics for mixed and hierarchical models*, PhD thesis, Los Angeles, University of California, 1995.
- [14] G. Ibacache-Pulgar, G.A. Paula, and M. Galea, *Influence diagnostics for elliptical semiparametric mixed models*, Stat. Model. 12 (2012), pp. 165–193.
- [15] J.D. Kloke, J.W. McKean, and M.M. Rashid, *Rank-based estimation and associated inferences for linear models with cluster correlated errors*, J. Amer. Statist. Assoc. 104 (2009), pp. 384–390.

- [16] J. Kowalsky, J.R. Mendoza-Blanco, X.M. Tu, and L.J. Gleser, *On the difference in inference and prediction between the joint and independent t-error models for seemingly unrelated regressions*, Comm. Statist. Theory Methods 28 (1999), pp. 2219–2140.
- [17] N.M. Laird, *Computation of variance components using the EM algorithm*, J. Stat. Comput. Simul. 14 (1982), pp. 295–303.
- [18] N.M. Laird and J.H. Ware, *Random-effects models for longitudinal data*, Biometrics 38 (1982), pp. 963–974.
- [19] K.L. Lange, R.J.A. Little, and M.G. Taylor, *Robust statistical modeling using the t distribution*, J. Amer. Statist. Assoc. 84 (1989), pp. 881–896.
- [20] J.K. Lindsey, *Multivariate elliptically contoured distributions for repeated measurements*, Biometrics 55 (1999), pp. 1277–1280.
- [21] R.J.A. Little, *Robust estimation of the mean and covariance matrix from data with missing values*, J. R. Stat. Soc. C 37 (1988), pp. 23–38.
- [22] C. Liu and D.B. Rubin, *ML estimation of the t distribution using EM and its extensions*, ECM and ECME, Statist. Sinica 5 (1995), pp. 19–39.
- [23] R.A. McLean, W.L. Sanders, and W.W. Stroup, *A unified approach to mixed linear models*, Amer. Statist. 45 (1991), pp. 54–64.
- [24] A.H. Munnell, *Why has productivity declined? Productivity and public investment*, N. Eng. Econom. Rev. (1990), pp. 3–22.
- [25] F. Osorio, G.A. Paula, and M. Galea, *Assessment of local influence in elliptical linear models with longitudinal structure*, Comput. Statist. Data Anal. 51 (2007), pp. 4354–4368.
- [26] J.C. Pinheiro and D.M. Bates, *Mixed-Effects Models in S and S-PLUS*, Springer-Verlag, Berlin, 2000.
- [27] J.C. Pinheiro, C. Liu, and Y.N. Wu, *Efficient algorithms for robust estimation in linear mixed-effects models using the multivariate t distribution*, J. Comput. Graph. Statist. 10 (2001), pp. 249–276.
- [28] S.W. Raudenbush and A.S. Bryk, *Hierarchical Linear Models: Applications and Data Analysis Methods*, Sage Publications, Thousand Oaks, 2002.
- [29] C. Savalli, G.A. Paula, and F.J.A. Cysneiros, *Assessment of variance components in elliptical linear mixed models*, Stat. Model. 6 (2006), pp. 59–76.
- [30] S.R. Searle, G. Casella, and C.E. McCulloch, *Variance Components*, Wiley, New York, 1992.
- [31] M.J. Silvapulle and P. Silvapulle, *A score test against one-sided alternatives*, J. Amer. Statist. Assoc. 90 (1995), pp. 342–349.
- [32] B.T. West, K.B. Welch, and A.T. Galecki, *Linear Mixed Models: A Practical Guide Using Statistical Software*, Chapman and Hall/CRC, Boca Raton, 2007.

Appendix. Three-level multilevel model notation

Considering P explanatory variables in the level 1, $\sum_{p=1}^P Q_p$ variables in the level 2 (where Q_p is the number of explanatory variables for the p th level 1 parameter) and $\sum_{p=1}^P \sum_{q=1}^{Q_p} S_{pq}$ variables in the level 3 (where S_{pq} is the number of variables associated with the q th level 2 parameter associated with the p th level 1 parameter), one has the general equations for the three-level multilevel model

$$\begin{aligned}
 \text{Level 1: } Y_{ijk} &= \sum_{p=1}^P \gamma_{pj|k} G_{pijk} + \epsilon_{ijk}; \\
 \text{Level 2: } \gamma_{pj|k} &= \sum_{q=1}^{Q_p} \delta_{pqk} T_{pqjk} + \xi_{pj|k}; \\
 \text{Level 3: } \delta_{pqk} &= \sum_{s=1}^{S_{pq}} \beta_{pqs} L_{pqsk} + \vartheta_{pqk},
 \end{aligned} \tag{A1}$$

where $i = 1, 2, \dots, n_{jk}$, $j = 1, 2, \dots, m_k$ and $k = 1, 2, \dots, K$. Here, n_{jk} is the total of individuals in the j th cluster in the level 2 and k th cluster in the level 3. One has also m_k as the total of clusters in the level 2 for the k th cluster of level 3 and K the total of clusters in the level 3. In Equation (A1) one has that Y_{ijk} is the response variable for the i th subject in the j th level 2 cluster and k th level 3 cluster, G_p , T_{pq} and L_{pqs} are explanatory variable values for the levels 1, 2 and 3, respectively, for $p = 1, 2, \dots, P$, $q = 1, 2, \dots, Q_p$ and $s = 1, 2, \dots, S_{pq}$, $\gamma_{pj|k}$, δ_{pqk} and β_{pqs} are parameters associated with the explanatory variable values G_p , T_{pq} and L_{pqs} , respectively, and ϵ_{ijk} , $\xi_{pj|k}$ and ϑ_{pqk} are random errors regarded to Y_{ijk} , $\gamma_{pj|k}$ and δ_{pqk} , respectively. Let

$$\begin{aligned} \mathbf{Y}_k &= (\mathbf{Y}_{1k}^\top, \mathbf{Y}_{2k}^\top, \dots, \mathbf{Y}_{m_k k}^\top)^\top_{(\sum_{j=1}^{m_k} n_{jk}) \times 1}, \quad \mathbf{Y}_{jk} = (Y_{1jk}, Y_{2jk}, \dots, Y_{n_{jk}jk})^\top_{n_{jk} \times 1}, \\ \mathbf{G}_k &= \text{diag}\{\mathbf{G}_{1k}, \mathbf{G}_{2k}, \dots, \mathbf{G}_{m_k k}\}_{(\sum_{j=1}^{m_k} n_{jk}) \times (Pm_k)}, \quad \mathbf{G}_{jk} = \{G_{pijk}\}_{n_{jk} \times P}, \\ \boldsymbol{\gamma}_k &= (\boldsymbol{\gamma}_{1k}^\top, \boldsymbol{\gamma}_{2k}^\top, \dots, \boldsymbol{\gamma}_{m_k k}^\top)^\top_{(Pm_k) \times 1}, \quad \boldsymbol{\gamma}_{jk} = (\gamma_{1j|k}, \gamma_{2j|k}, \dots, \gamma_{Pj|k})^\top_{P \times 1}, \\ \boldsymbol{\epsilon}_k &= (\boldsymbol{\epsilon}_{1k}^\top, \boldsymbol{\epsilon}_{2k}^\top, \dots, \boldsymbol{\epsilon}_{m_k k}^\top)^\top_{(\sum_{j=1}^{m_k} n_{jk}) \times 1}, \quad \boldsymbol{\epsilon}_{jk} = (\epsilon_{1jk}, \epsilon_{2jk}, \dots, \epsilon_{n_{jk}jk})^\top_{n_{jk} \times 1}, \\ \mathbf{T}_k &= [\mathbf{T}_{1k}^\top | \mathbf{T}_{2k}^\top | \dots | \mathbf{T}_{m_k k}^\top]^\top_{(Pm_k) \times (\sum_{p=1}^P Q_p)}, \\ \mathbf{T}_{jk} &= \text{diag}\{\mathbf{T}_{1jk}, \mathbf{T}_{2jk}, \dots, \mathbf{T}_{Pjk}\}_{P \times \sum_{p=1}^P Q_p}, \quad \mathbf{T}_{pjk} = (T_{p1jk}, T_{p2jk}, \dots, T_{pQ_pjk})_{1 \times Q_p}, \\ \boldsymbol{\delta}_k &= (\delta_{11k}, \delta_{12k}, \dots, \delta_{1Q_1k}, \dots, \delta_{P1k}, \delta_{P2k}, \dots, \delta_{PQ_Pk})^\top_{(\sum_{p=1}^P Q_p) \times 1}, \\ \mathbf{L}_k &= \text{diag}\{\mathbf{L}_{11k}, \mathbf{L}_{12k}, \dots, \mathbf{L}_{1Q_1k}, \dots, \mathbf{L}_{P1k}, \mathbf{L}_{P2k}, \dots, \mathbf{L}_{PQ_Pk}\}_{(\sum_{p=1}^P Q_p) \times (\sum_{p=1}^P \sum_{q=1}^{Q_p} S_{pq})}, \\ \mathbf{L}_{pqk} &= (L_{pq1k}, L_{pq2k}, \dots, L_{pqS_{pq}k})_{1 \times S_{pq}}, \\ \boldsymbol{\beta} &= (\beta_{111}, \beta_{112}, \dots, \beta_{11S_{11}}, \dots, \beta_{PQ_P1}, \beta_{PQ_P2}, \dots, \beta_{PQ_P S_{PQ_P}})^\top_{(\sum_{p=1}^P \sum_{q=1}^{Q_p} S_{pq}) \times 1}, \\ \boldsymbol{\xi}_k &= (\boldsymbol{\xi}_{1k}^\top, \boldsymbol{\xi}_{2k}^\top, \dots, \boldsymbol{\xi}_{m_k k}^\top)^\top_{(Pm_k) \times 1}, \quad \boldsymbol{\xi}_{jk} = (\xi_{1j|k}, \xi_{2j|k}, \dots, \xi_{Pj|k})^\top_{P \times 1} \end{aligned}$$

and

$$\boldsymbol{\vartheta}_k = (\vartheta_{11k}, \vartheta_{12k}, \dots, \vartheta_{1Q_1k}, \dots, \vartheta_{P1k}, \vartheta_{P2k}, \dots, \vartheta_{PQ_Pk})^\top_{(\sum_{p=1}^P Q_p) \times 1}.$$

Therefore, the multilevel model for the k th cluster is given by

$$\mathbf{Y}_k = \mathbf{G}_k \boldsymbol{\gamma}_k + \boldsymbol{\epsilon}_k; \quad (\text{A2})$$

$$\boldsymbol{\gamma}_k = \mathbf{T}_k \boldsymbol{\delta}_k + \boldsymbol{\xi}_k; \quad (\text{A3})$$

$$\boldsymbol{\delta}_k = \mathbf{L}_k \boldsymbol{\beta} + \boldsymbol{\vartheta}_k. \quad (\text{A4})$$

Replacing Equation (A4) in Equation (A3) and Equation (A3) in Equation (A2), one obtains

$$\mathbf{Y}_k = \mathbf{G}_k [\mathbf{T}_k (\mathbf{L}_k \boldsymbol{\beta} + \boldsymbol{\vartheta}_k) + \boldsymbol{\xi}_k] + \boldsymbol{\epsilon}_k = \mathbf{G}_k \mathbf{T}_k \mathbf{L}_k \boldsymbol{\beta} + \mathbf{G}_k \mathbf{T}_k \boldsymbol{\vartheta}_k + \mathbf{G}_k \boldsymbol{\xi}_k + \boldsymbol{\epsilon}_k.$$

Thus, one has $\mathbf{Y}_k = \mathbf{X}_k \boldsymbol{\beta} + \mathbf{Z}_k \mathbf{b}_k + \boldsymbol{\epsilon}_k$, where $\mathbf{X}_k = \mathbf{G}_k \mathbf{T}_k \mathbf{L}_k$, $\mathbf{Z}_k = [\mathbf{G}_k \mathbf{T}_k | \mathbf{G}_k]$ and $\mathbf{b}_k = (\boldsymbol{\vartheta}_k^\top, \boldsymbol{\xi}_k^\top)^\top$. In case of some parameters in $\boldsymbol{\gamma}_k$ and/or in $\boldsymbol{\delta}_k$ being considered as fixed, the random-effects vector $\mathbf{b}_k$ must be properly arranged by omitting the errors related to the respective parameters, and the columns of $\mathbf{Z}_k$ related to the omitted errors are suppressed. Considering the K level 3 clusters, one

has

$$\begin{aligned}
\mathbf{Y} &= (\mathbf{Y}_1^\top, \mathbf{Y}_2^\top, \dots, \mathbf{Y}_K^\top)^\top_{(\sum_{k=1}^K \sum_{j=1}^{m_k} n_{jk}) \times 1}, \\
\mathbf{G} &= \text{diag}\{\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_K\}_{(\sum_{k=1}^K \sum_{j=1}^{m_k} n_{jk}) \times (P \sum_{k=1}^K m_k)}, \\
\boldsymbol{\gamma} &= (\boldsymbol{\gamma}_1^\top, \boldsymbol{\gamma}_2^\top, \dots, \boldsymbol{\gamma}_K^\top)^\top_{(P \sum_{k=1}^K m_k) \times 1}, \quad \boldsymbol{\epsilon} = (\boldsymbol{\epsilon}_1^\top, \boldsymbol{\epsilon}_2^\top, \dots, \boldsymbol{\epsilon}_K^\top)^\top_{(\sum_{k=1}^K \sum_{j=1}^{m_k} n_{jk}) \times 1}, \\
\mathbf{T} &= \text{diag}\{\mathbf{T}_1, \mathbf{T}_2, \dots, \mathbf{T}_K\}_{(P \sum_{k=1}^K m_k) \times (K \sum_{p=1}^P Q_p)}, \\
\boldsymbol{\delta} &= (\boldsymbol{\delta}_1^\top, \boldsymbol{\delta}_2^\top, \dots, \boldsymbol{\delta}_K^\top)^\top_{(K \sum_{p=1}^P Q_p) \times 1}, \quad \boldsymbol{\xi} = (\boldsymbol{\xi}_1^\top, \boldsymbol{\xi}_2^\top, \dots, \boldsymbol{\xi}_K^\top)^\top_{(P \sum_{k=1}^K m_k) \times 1}, \\
\mathbf{L} &= [\mathbf{L}_1^\top | \mathbf{L}_2^\top | \dots | \mathbf{L}_K^\top]^\top_{(K \sum_{p=1}^P Q_p) \times (\sum_{p=1}^P \sum_{q=1}^{Q_p} s_{pq})},
\end{aligned}$$

and

$$\boldsymbol{\vartheta} = (\boldsymbol{\vartheta}_1^\top, \boldsymbol{\vartheta}_2^\top, \dots, \boldsymbol{\vartheta}_K^\top)^\top_{(K \sum_{p=1}^P Q_p) \times 1}.$$

Therefore, one has the multilevel model for the K clusters given by

$$\mathbf{Y} = \mathbf{G}\boldsymbol{\gamma} + \boldsymbol{\epsilon}; \tag{A5}$$

$$\boldsymbol{\gamma} = \mathbf{T}\boldsymbol{\delta} + \boldsymbol{\xi}; \tag{A6}$$

$$\boldsymbol{\delta} = \mathbf{L}\boldsymbol{\beta} + \boldsymbol{\vartheta}. \tag{A7}$$

Replacing Equation (A7) in Equation (A6) and Equation (A6) in Equation (A5), one obtains

$$\mathbf{Y} = \mathbf{G}[\mathbf{T}(\mathbf{L}\boldsymbol{\beta} + \boldsymbol{\vartheta}) + \boldsymbol{\xi}] + \boldsymbol{\epsilon} = \mathbf{GTL}\boldsymbol{\beta} + \mathbf{GT}\boldsymbol{\vartheta} + \mathbf{G}\boldsymbol{\xi} + \boldsymbol{\epsilon}.$$

Thus, one has $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \boldsymbol{\epsilon}$, where $\mathbf{X} = \mathbf{GTL}$, $\mathbf{Z} = [\mathbf{GT} | \mathbf{G}]$ and $\mathbf{b} = (\boldsymbol{\vartheta}^\top, \boldsymbol{\xi}^\top)^\top$.